

# Introductory Econometrics with Recitation: TA Session Week 14

Danny Po-Hsien Kang (康柏賢)

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# Today's agenda

- 1 Instrumental Variable (IV) Regression
  - Story and Model Setup
  - OLS and IV Model
- 2 IV Validity
  - Relevance
  - Exogeneity
- 3 Concluding Remark
  - Acknowledgement

# Final

- Final: 9:30 AM - 12:00 PM, June 3 (Wed)
- You can bring: One cheat sheet, Calculator
- No questions about R will appear in the final exam!
- No TA Session in final week.
  - TA office hour: June 2 (Tue) 13:20-17:20

# Today's Dataset

- `smoke_ban.csv`: a cross-sectional dataset on smoking behavior and income from the `wooldridge` package.
- Import the data:

```
smoke <- wooldridge::smoke
```

- The main question is whether cigarette smoking affects income.

| Variable              | Description                                        |
|-----------------------|----------------------------------------------------|
| <code>lnincome</code> | Log of annual income                               |
| <code>cigs</code>     | Number of cigarettes smoked per day                |
| <code>cigpric</code>  | State cigarette price, measured in cents per pack. |
| <code>restaurn</code> | Restaurant smoking ban; 1 = Ban, 0 = No ban        |
| <code>educ</code>     | Years of education                                 |
| <code>age</code>      | Age in years                                       |
| <code>agesq</code>    | Age squared                                        |

# Story: Costly Cigarettes

- Smoking cigarettes is not only addictive, but also costly.
- We want to study the effect of smoking on income.
- Consider the following model:

$$\text{lincome}_i = \beta_0 + \beta_1 \text{cigs}_i + \beta_2 \text{educ}_i + \beta_3 \text{age}_i + \beta_4 \text{agesq}_i + u_i.$$

- The variable `cigs` may be endogenous, so  $\beta_1$  may not have a causal interpretation.

# Story: Costly Cigarettes

- Some states in the U.S. have implemented smoking bans in restaurants.
- This provides a possible instrumental variable: `restaurn`.
- The IV idea requires two conditions:
  - **Relevance:** restaurant bans affect smoking behavior.
  - **Exogeneity:** restaurant bans affect income only through smoking behavior.
- We use two-stage least squares (2SLS) to estimate the IV model.

# Two-Stage Least Squares

- **First stage:**

$$cigs_i = \pi_0 + \pi_1 restaurn_i + \pi_2 educ_i + \pi_3 age_i + \pi_4 agesq_i + v_i.$$

- **Second stage:**

$$lincome_i = \beta_0 + \beta_1 \widehat{cigs}_i + \beta_2 educ_i + \beta_3 age_i + \beta_4 agesq_i + u_i.$$

- Here,  $\widehat{cigs}_i$  is the predicted value from the first-stage regression.

# OLS and IV Model

- **OLS model**

```
ols <- lm_robust(  
  lincome ~ cigs + educ + age + agesq,  
  data = smoke  
)
```

- **IV model**

```
library(ivreg)  
  
iv <- ivreg(  
  lincome ~ cigs + educ + age + agesq |  
    restaurn + educ + age + agesq,  
  data = smoke  
)
```



# Model Comparison

## • Output:

|             | OLS                     | IV                      |
|-------------|-------------------------|-------------------------|
| (Intercept) | 7.7954 ***<br>(0.2085)  | 7.7811 ***<br>(0.2281)  |
| cigs        | 0.0017<br>(0.0014)      | -0.0414<br>(0.0260)     |
| educ        | 0.0604 ***<br>(0.0075)  | 0.0400 *<br>(0.0162)    |
| age         | 0.0577 ***<br>(0.0092)  | 0.0932 ***<br>(0.0237)  |
| agesq       | -0.0006 ***<br>(0.0001) | -0.0010 ***<br>(0.0003) |
| $R^2$       | 0.1650                  | -0.4941                 |
| Adj. $R^2$  | 0.1608                  | -0.5015                 |
| Num. obs.   | 807                     | 807                     |
| RMSE        | 0.6529                  |                         |

\*\*\* p < 0.001; \*\* p < 0.01; \* p < 0.05

# IV Estimation with Robust SE

- Use `coeftest()` with a robust variance-covariance estimator.

```
library(lmtest)    # for coeftest()
library(sandwich)  # for robust SE

coeftest(iv, vcov = sandwich)
```

## Output:

t test of coefficients:

|                | Estimate    | Std. Error | t value | Pr(> t )  |      |     |      |     |     |     |   |
|----------------|-------------|------------|---------|-----------|------|-----|------|-----|-----|-----|---|
| (Intercept)    | 7.78113928  | 0.25650420 | 30.3353 | < 2.2e-16 | ***  |     |      |     |     |     |   |
| cigs           | -0.04138462 | 0.02403335 | -1.7220 | 0.085461  | .    |     |      |     |     |     |   |
| educ           | 0.04002417  | 0.01536913 | 2.6042  | 0.009379  | **   |     |      |     |     |     |   |
| age            | 0.09320769  | 0.02230429 | 4.1789  | 3.251e-05 | ***  |     |      |     |     |     |   |
| agesq          | -0.00104373 | 0.00025489 | -4.0948 | 4.655e-05 | ***  |     |      |     |     |     |   |
| ---            |             |            |         |           |      |     |      |     |     |     |   |
| Signif. codes: | 0           | '***'      | 0.001   | '**'      | 0.01 | '*' | 0.05 | '.' | 0.1 | ' ' | 1 |

## Relevance: First-Stage F Test

- The instrument should be correlated with the endogenous variable.
- We check this using the first-stage regression.

```
first <- lm_robust(  
  cigs ~ restaurn + educ + age + agesq,  
  data = smoke  
)  
  
summary(first)
```

- A common rule of thumb is that the first-stage F-statistic should be larger than 10.

# Relevance: First-Stage F Test

- **Output:**

```
... (omitted)
```

```
Multiple R-squared: 0.0418
```

```
Adjusted R-squared: 0.03822
```

```
F-statistic: 16.46 on 3 and 803 DF
```

```
p-value: 2.176e-10
```

- Since the F-statistic is larger than 10, the instrument is not weak by this rule of thumb.

# Exogeneity: Overidentification Test

- If we have more instruments than endogenous variables, we can use the J test to test instrument exogeneity.
- After estimating the IV model, regress the 2SLS residual on the instruments:

$$\hat{u}_i^{2SLS} = \gamma_0 + \gamma_1 Z_{1i} + \gamma_2 Z_{2i} + \cdots + \gamma_m Z_{mi} + \epsilon_i.$$

- The null hypothesis is that all instruments are exogenous:

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_m = 0.$$

# Exogeneity: Overidentification Test

- The test statistic is:

$$J = nR^2 \sim \chi_{m-k}^2,$$

where  $m$  is the number of instruments and  $k$  is the number of endogenous variables.

- If we reject the null, then at least one instrument may be invalid.
- If we fail to reject the null, then the test does not find evidence against instrument exogeneity.
- This test requires overidentification, so it cannot be used when there is only one instrument for one endogenous variable.

# J-Test in R

- To run a J-test, suppose we use two instruments: restaurn and cigpric:

```
iv_overid <- ivreg(  
  lincome ~ cigs + educ + age + agesq |  
    restaurn + cigpric + educ + age + agesq,  
  data = smoke  
)  
summary(iv_overid, diagnostics = TRUE)
```

# J-Test in R

## • Output:

... (omitted)

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 7.7812628  | 0.2272624  | 34.239  | < 2e-16 ***  |
| cigs        | -0.0410122 | 0.0259272  | -1.582  | 0.114085     |
| educ        | 0.0401998  | 0.0160987  | 2.497   | 0.012721 *   |
| age         | 0.0929009  | 0.0235875  | 3.939   | 8.91e-05 *** |
| agesq       | -0.0010402 | 0.0002713  | -3.834  | 0.000136 *** |

Diagnostic tests:

|                  | df1 | df2 | statistic | p-value  |
|------------------|-----|-----|-----------|----------|
| Weak instruments | 2   | 801 | 3.131     | 0.0442 * |
| Wu-Hausman       | 1   | 801 | 4.887     | 0.0273 * |
| Sargan           | 1   | NA  | 1.519     | 0.2177   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.87 on 802 degrees of freedom

Multiple R-Squared: -0.4827, Adjusted R-squared: -0.4901

Wald test: 22.79 on 4 and 802 DF, p-value: < 2.2e-16



# Acknowledgement

- The structure of slides is based on the previous year TA course by Xiang-Jyun Jhang.
- This is the last TA session, and next week will be office hour: 6/2 (Tue) 13:20-17:20.
  - Thank you for joining the TA sessions this year!
  - I hope the TA sessions were helpful.
- If you have questions or just want to chat, my door is always open.
- You can reach me by email or in person.
  - Course suggestions
  - Econometrics and empirical research
  - Graduate school preparation and applications

# Suggested Courses for Future Learning

- **Econometrics I and II**

- Revisit concepts from this semester with vector and matrix notation.
- Introduce more empirical methods and models.
- Usually require an empirical research project.

- **Econometrics Theory I and II**

- Extend econometric models using linear algebra.
- Focus on both theory and applications.
- Cover asymptotic theory, panel data, and potential outcomes.